EXAM STUDY COMPENDIUM & THEORY GUIDE

Unit II: Ethical Decision-Making, Responsibility & Design for Values

Course Module: *Unit II — Ethical Decision-Making in AI, Normative Theories, Values in Practice, Implementing Ethical Reasoning, Responsible Research & Innovation, Accountability, Transparency, and Design for Values* | ***Textbook References:*** *S. M. Liao (Intro), F. M. Kamm (Ch. 2), J.-F. Bonnefon et al. (Ch. 3), A. Loreggia et al. (Ch. 4), P. Asaro (Ch. 7), C. O'Neil & H. Gunn (Ch. 8), S. Wolfram (Ch. 5).*

UNIT II SYLLABUS ROADMAP & CORE THEMES

Unit II covers the translation of normative ethics into algorithmic practice. Mastery is required across five key pillars: (1) Philosophical comparison of normative theories (Utilitarianism, Deontology, Virtue Ethics) and the limits of direct algorithmic translation; (2) Frances Kamm's analysis of the Trolley Problem in autonomous vehicles, distinguishing threat redirection from new threat introduction, and evaluating driver vs. pedestrian liability; (3) The moral psychology of AI, the *Ethical Opt-Out Problem*, and the Social Dilemma of Autonomous Vehicles (Bonnefon, Shariff, Rahwan); (4) Computational formalisms for ethics: hard vs. soft constraints, Conditional Preference networks (CP-nets), distance metrics (I-CPD), and alignment thresholds (Loreggia et al.); and (5) Design for values, algorithmic accountability, and transparency via the *Ethical Matrix* (O'Neil & Gunn) and meaningful human control in autonomous weapons (Asaro).

1. Normative Ethical Theories in Artificial Intelligence

Ethical decision-making in autonomous systems requires understanding the three dominant Western normative ethical traditions. Each theory provides a distinct decision criterion for moral evaluation, and each introduces unique mathematical and conceptual challenges when implemented in machines:

Ethical Theory	Foundational Premise & Decision Criterion	Implementation in AI Systems	Critical Failures & Bottlenecks in AI
Utilitarianism / Consequentialism (Bentham, Mill, Singer)	An action is morally right if and only if it maximizes aggregate net welfare or expected utility across all affected parties (*greatest good for greatest number*).	Modeled as cost minimization or objective function optimization (e.g., minimizing expected traffic fatalities).	**Calculation Problem:** Intractable combinatorial branches; unknown future probabilities; Black Swan events; justifies trampling minority rights to benefit the majority.
Deontology / Duty Ethics (Kant, Ross, Rawls)	Actions are intrinsically right or wrong based on duties, moral laws, and rights, regardless of consequences. Respect for human dignity (Kantian Categorical Imperative: treat persons as ends, never mere means).	Modeled via **hard constraints**, deontic logic, and priority rules where forbidden actions are excluded from the solution space.	**Algorithmic Brittleness:** Absolute prohibitions cannot resolve conflicting duties; cannot handle probabilistic risk trade-offs gracefully; risks paralysis in dilemmas.

Virtue Ethics (Aristotle, Vallor, Wallach)	Ethics is grounded in moral character and cultivated virtues (wisdom, justice, courage, temperance, benevolence) guided by practical wisdom (*phronesis*).	Bottom-up reinforcement learning from moral exemplars; context-adaptive character cultivation; embodied virtue navigation.	**Formalization Void:** Virtues resist reduction to static utility scalars; requires holistic world understanding, moral perception, and affective sensitivity.

1.1 Why Directly Hard-Coding Overarching Theories Fails (Liao):
Liao points out three major obstacles to top-down moral programming: (1) *Epistemic calculation problems:* In utilitarianism, an algorithm cannot compute all secondary and tertiary ripple effects of an action. (2) *Normative disagreement:* Society lacks consensus on which theory is correct. Imposing Kantian deontology or Millian utilitarianism as universal software encodes contentious philosophical dogmas. (3) *Rule satisfaction ambiguity:* Top-down rules fail to specify when their own application criteria have been satisfied in messy, unstructured environments.

2. Values in Practice: Kamm's Deconstruction of the Trolley Problem

In Chapter 2 (*The Use and Abuse of the Trolley Problem*), **F. M. Kamm** provides an exacting philosophical critique of the popular tendency to treat autonomous vehicle crashes as simple Trolley Problems. Kamm establishes vital conceptual and moral distinctions:

2.1 Redirecting an Existing Threat vs. Introducing a New Threat:
In the classic *Switch Case* (Philippa Foot; Judith Thomson), an existing threat (runaway trolley) is already hurtling toward five people, and an agent diverts that very threat onto a sidetrack where one person dies. The action is widely judged permissible under the *Doctrine of Double Effect (DDE)*: the harm to the one is a foreseen side effect, not an intended means. In contrast, in the *Footbridge / Topple Case* or *Bomb Case*, the agent introduces a **new harmful threat** (pushing a heavy man or detonating a bomb) to stop the trolley. This is strictly impermissible because the victim is used as a mere means. *Application to AVs:* The **Ambulance Case** illustrates this. If an ambulance is carrying five dying patients to a hospital, it is morally impermissible to program the ambulance to run over one innocent pedestrian to get the patients to the hospital on time. The ambulance running over the pedestrian introduces an independent, new threat to someone not already threatened by the disease.

2.2 Threateners vs. Non-Threateners (Innocent Threat Cases):
In standard Trolley cases, the decision-maker (driver or bystander) is completely safe and chooses between nonthreatening third parties. In real-world car crashes (e.g., Rampolla's skidding car scenario), the driver *is the threat*. This is an **Innocent Threat Case**: someone who is presenting a deadly threat to others (even without moral fault) must decide whether to harm innocent pedestrians or allow harm to themselves. Thomson argued that a driver setting a lethal threat in motion has a strict duty to divert into a barrier, sacrificing themselves, rather than strike innocent pedestrians.

2.3 Distribution of Risk and Degrees of Liability:
Kamm argues that algorithms cannot simply minimize raw body counts; they must account for moral **liability to bear harm**:

- **The Irresponsible Five Case:** Suppose five pedestrians illegally sprint onto an active highway against a red light, and the car can only avoid them by swerving into an innocent person walking on a sidewalk. Kamm argues the car should *not* swerve. Because the five assumed the risk through their wrongful action, they possess higher liability to bear the loss. Liability constrains or lexicographically overrides pure headcount utilitarianism.

• **'Playing the Game of Driving':** When an AV must choose between colliding with another moving car (with a driver/passengers) or hitting a pedestrian on a sidewalk, Kamm suggests the car should at least sometimes redirect toward the other car. Motorists voluntarily participate in the dangerous activity of motorized transit, whereas pedestrians on sidewalks are non-participants who have not assumed road hazards.

• **Programmers as Intrusive Bystanders:** In partial autonomy cases, an algorithmic system that seizes control from a driver and forces the car into a fatal collision behaves like an 'intrusive bystander' taking liberty away from a moral agent. Commercial manufacturers cannot sell 'immoral cars' programmed to sacrifice innocent pedestrians to protect selfish buyers simply to maximize sales.

3. Moral Psychology of AI & The Ethical Opt-Out Problem

In Chapter 3, **Jean-François Bonnefon, Azim Shariff, and Iyad Rahwan** examine public attitudes and experimental psychology surrounding autonomous machines. They formulate the **Ethical Opt-Out Problem**: even if moral philosophers reach consensus on an ideal ethical algorithm, their work is rendered null and void if citizens strongly disagree with the underlying trade-offs and refuse to adopt the technology.

THE SOCIAL DILEMMA OF AUTONOMOUS VEHICLES

The Social Dilemma of Autonomous Vehicles (Bonnefon et al., Science 2016):
Through large-scale psychological surveys, researchers uncovered a profound social dilemma: (1) When evaluating AV policy from a societal perspective behind a 'veil of ignorance', citizens overwhelmingly favor **utilitarian AVs** that minimize total casualties (even if the AV sacrifices its passenger to save multiple pedestrians). (2) However, when asked what kind of car they would purchase for themselves and their families, citizens overwhelmingly refuse to buy utilitarian AVs, demanding **self-protective AVs**. (3) If governments legally mandate utilitarian algorithms, consumers will simply **opt out of buying AVs**, maintaining human-driven vehicles. Because human drivers cause vastly more fatal accidents than autonomous systems, this ethical opt-out paradoxically results in *more overall road fatalities*.

3.2 Stylized Dilemmas vs. Statistical Risk Distribution (The 'Bull Bar' Analogy):
Critics often attack Trolley problems as unrealistic edge cases that rarely occur. Bonnefon et al. defend stylized dilemmas as indispensable models: In everyday operation, an AV's software makes micro-maneuvers that subtly shift statistical probabilities of harm between passengers, cyclists, and pedestrians. Aggregated over millions of miles, these statistical shifts become visible as fatal disparities (e.g., Car A having a 1:2 passenger-to-pedestrian death ratio vs. Car B having a 1:7 ratio). Just as regulators banned 'bull bars' (metal grilles) on SUVs because they protected occupants while crushing pedestrians, society must regulate the ethical trade-offs embedded in autonomous vehicle software.

4. Implementing Ethical Reasoning: Constraints & CP-Nets (Loreggia et al.)

In Chapter 4, **Andrea Loreggia, Nicholas Mattei, Francesca Rossi, and K. Brent Venable** provide a computational framework for modeling subjective preferences alongside exogenous ethical priorities using constraint processing and graphical preference models:

4.1 Hard vs. Soft Constraints in Machine Ethics:
• **Hard Constraints (Deontology):** Variables represent action attributes; constraints model ethical prohibitions. An

action is *permissible* if it belongs to the solution set; *impermissible* if it violates constraints; and *obligatory* if it is the sole solution.

• **Soft / Weighted Constraints (Utilitarianism):** Assigns cost or reward weights to attribute combinations. Translates utility maximization into cost minimization over semiring structures. Fuzzy constraints use the *min* operator, ensuring that a violation of any ethical requirement degrades the overall solution.

4.2 Conditional Preference Networks (CP-Nets):

A CP-net is a compact graphical formalism capturing qualitative, ceteris paribus ('all else being equal') preference statements. It consists of a *directed dependency graph* (showing which features influence preferences over other features) and a collection of *Conditional Preference Tables (CP-tables)*. For example, in an AV dilemma with features: *Hurt Entities (H: Cat vs. Person)*, *Number of Victims (N: Few vs. Many)*, and *Direction (D: Straight vs. Swerve)*, a CP-net induces a partial order over all 8 combinatorial outcomes via sequence of **worsening flips**.

4.3 The Value Alignment Procedure via CP-Net Distance:

Loreggia et al. propose measuring the distance between an agent's subjective CP-net (N_p) and the community's ethical CP-net (N_e) using tractable metrics (such as the **I-CPD distance**):

1. **Define Thresholds:** Set threshold $t_1 \in [0, 1]$ (maximum allowable divergence between preference graphs) and $t_2 \in [1, n]$ (maximum allowable worsening flips between the agent's action and the community's optimal action).
2. **Distance Evaluation:** If $dist(N_p, N_e) \le t_1$, the agent is deemed aligned and executes its top preferred action.
3. **Compromise Guidance:** If $dist(N_p, N_e) > t_1$, the agent is unaligned. The system forces the agent to traverse down its induced preference order via worsening flips until it selects an action within t_2 flips of the community's optimal outcome.

5. Accountability, Transparency, and Design for Values

In Chapter 8, **Cathy O'Neil and Hanna Gunn** address near-term algorithmic injustice in public systems. They argue that algorithmic developers and deployers make profound moral choices disguised as technical engineering parameters. To establish genuine accountability, they propose adapting **Ben Mepham's Ethical Matrix**:

Stakeholder Group	Respect for Well-Being (Beneficence & Non-maleficence)	Respect for Autonomy & Dignity (Freedom & Agency)	Respect for Justice & Fairness (Rawlsian Equity & Rights)
Deployers / Producers (Government / Corporation)	System efficiency, administrative cost reduction, organizational stability.	Freedom to adopt innovative technology; protection of intellectual property.	Fraud reduction; contract compliance; legal liability insulation.
Evaluated Targets (Welfare Clients, Defendants)	Receipt of vital necessities (food, healthcare, housing); freedom from wrongful detention.	Right to transparent explanation of own case; meaningful informed consent; human appeal.	Equal protection; non-discrimination; due process; balanced false-positive/negative error rates.
General Public	Public safety; economic flourishing; maintenance of social trust.	Civic transparency; democratic oversight over algorithmic governance.	Fair distribution of public burdens; elimination of systemic bias in institutions.
Biota / Environment	Ecological health; minimization of environmental degradation.	Maintenance of natural habitats and biodiversity.	Sustainable intergenerational resource allocation.

5.1 Case Studies in Algorithmic Harm:

• **Indiana FSSA Welfare Automation (Virginia Eubanks):** Indiana contracted IBM/ACS to automate welfare eligibility, aiming to cut rolls and reduce fraud. The automated system sent blanket notices citing 'failure to cooperate' whenever documents were misplaced, denying Medicaid and food stamps to thousands of disabled and vulnerable clients without appeal. *Ethical Failure:* The design team considered only the state's narrow financial metrics while completely ignoring clients' well-being, autonomy, and procedural justice.

• **COMPAS Recidivism Risk Scoring:** Used across US courts to advise bail and sentencing. Northpointe defended COMPAS by claiming *predictive parity* (a score of 7 meant equal probability of re-arrest across races). However, ProPublica revealed that among defendants who did *not* re-offend, Black defendants were twice as likely to be falsely classified as high risk (false positives), while White defendants were misclassified as low risk (false negatives). *Ethical Failure:* Selecting a single statistical metric masks fundamental ethical trade-offs between accuracy equity and civil liberties.

5.2 Meaningful Human Control in Autonomous Weapons (Peter Asaro, Ch. 7):

Peter Asaro demonstrates why lethal decisions cannot be delegated to autonomous weapon systems (LAWS). Target discrimination is not moral recognition: classifying an object in a sensor array does not respect human dignity. Under international humanitarian law, lethal force requires: (1) understanding the value and irreversibility of human life; (2) assessing military necessity and proportionality; and (3) taking moral and legal responsibility. Because computational systems lack moral agency, delegating lethal decisions creates a fatal **Responsibility Gap** (Matthias), necessitating mandatory *Meaningful Human Control*.

6. High-Probability Exam Questions & Model Answer Outline

Q1: Contrast Frances Kamm's Ambulance Case with the standard Trolley Switch Case. Why does this distinction matter for programming autonomous vehicles?

Model Outline: (1) In the Switch Case, an existing threat is redirected to save five while foreseeably killing one (justified under DDE as non-intended side effect). (2) In the Ambulance Case, patients are dying of an internal medical condition; running over a pedestrian introduces a new, independent threat to an innocent person who was not part of the original danger. (3) Apply to AVs: Programmers cannot justify running over pedestrians on the route simply because doing so saves a greater number of vehicle passengers. Redirection is morally distinct from creating new casualties.

Q2: Explain the Ethical Opt-Out problem and the Social Dilemma of Autonomous Vehicles formulated by Bonnefon, Shariff, and Rahwan.

Model Outline: (1) Define Ethical Opt-Out: when algorithmic ethical trade-offs conflict with citizen moral preferences, users opt out of the system. (2) Detail the AV Social Dilemma: citizens want utilitarian AVs for everyone else (veil of ignorance) but self-protective AVs for their own purchase. (3) Explain the regulatory paradox: mandating utilitarian AVs leads to consumer boycotts, leaving human drivers on the road and increasing overall traffic fatalities.